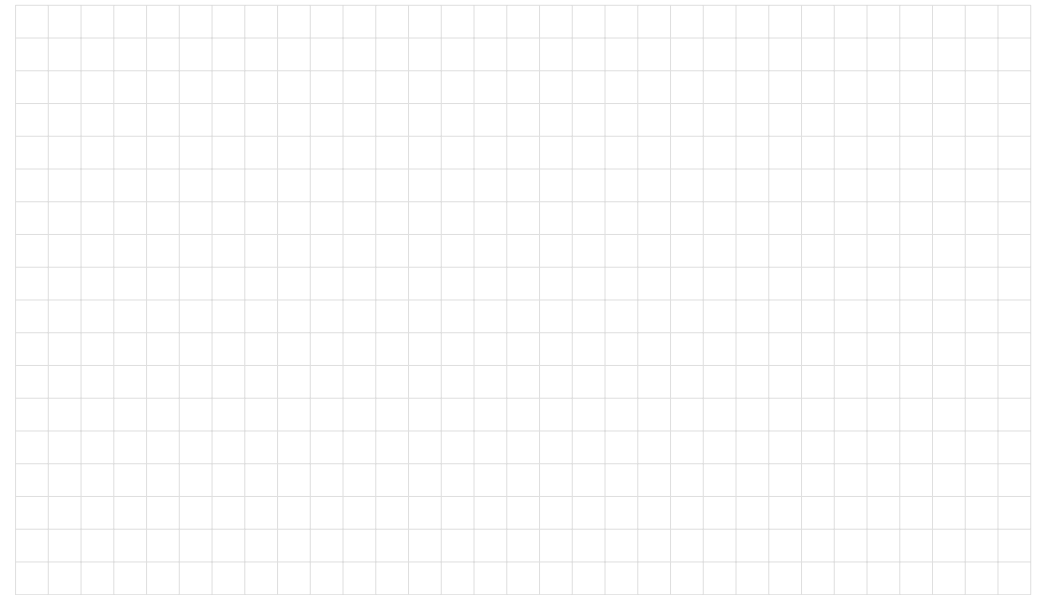


Elements of Matrix Theory

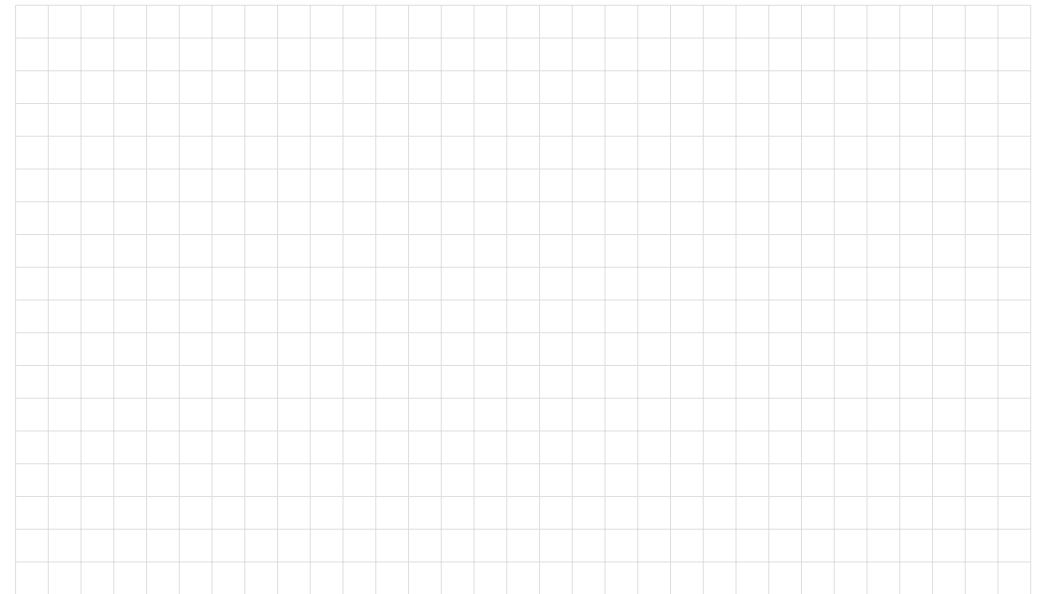
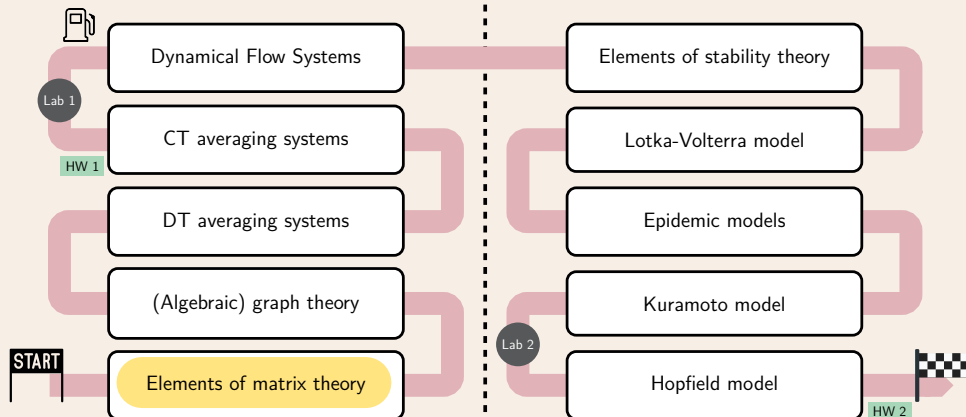
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Network Dynamical Systems
Master Degree in Control Systems Engineering
A.A. 2025-2026



Linear Dynamics

Nonlinear Dynamics



Covered topics

- ▷ DT linear systems and Jordan form
- ▷ Stochastic matrices
- ▷ Perron-Frobenius theory
- ▷ Applications to DT averaging systems



Standard notation

- ▷ $\mathbf{1}_n$ = n -dimensional vector of all ones
- ▷ $\mathbf{0}_{n \times m}$ = $n \times m$ matrix of all zeros
- ▷ $\mathbf{e}_i$ = i -th vector of the canonical basis of $\mathbb{R}^n$
- ▷ $\mathbf{x} \in \mathbb{R}^n$, $\|\mathbf{x}\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$
- ▷ $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{A}^\top \in \mathbb{R}^{m \times n}$ = transpose of $\mathbf{A}$
- ▷ $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\text{spec}(\mathbf{A})$ = spectrum (set of eigenvalues) of $\mathbf{A}$
- ▷ $\mathbf{A} \succ 0$ ($\mathbf{A} \succeq 0$) = $\mathbf{A}$ positive definite (positive semi-definite)
- ▷ $\ker(\mathbf{A})$ = kernel of $\mathbf{A}$, $\text{im}(\mathbf{A})$ = image of $\mathbf{A}$, $\text{rank}(\mathbf{A})$ = rank of $\mathbf{A}$
- ▷ $\{\mathbf{v}_i\}_{i=1}^k$, $\mathbf{v}_i \in \mathbb{R}^n$, $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ = subspace of $\mathbb{R}^n$ spanned by $\{\mathbf{v}_i\}_{i=1}^k$

$$\mathbf{1}_n = \overbrace{[1 \dots 1]}^n \quad \mathbf{e}_i = [0 \dots \underbrace{1}_i \dots 0]^\top$$

$$\mathbf{A} \succ 0 : \mathbf{A} = \mathbf{A}^\top \text{ \& } \mathbf{x}^\top \mathbf{A} \mathbf{x} > 0 \quad \forall \mathbf{x} \neq 0$$

$$\mathbf{A} \succeq 0 : \mathbf{A} = \mathbf{A}^\top \text{ \& } \mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0 \quad \forall \mathbf{x} \neq 0$$

DT linear systems and convergence

In the first part of the course, we will consider discrete-time, autonomous, linear time-invariant systems:

$$x(k+1) = Ax(k), \quad k \in \mathbb{Z}_{\geq 0}, \quad x(0) = x_0 \in \mathbb{R}^n$$

whose solution is

$$x(k) = A^k x_0, \quad k \in \mathbb{Z}_{\geq 0}.$$

Def: A matrix $A \in \mathbb{R}^{n \times n}$ is

(i) semi-convergent if $\lim_{k \rightarrow +\infty} A^k$ exists (finite);

(ii) convergent (or Schur stable) if it is semi-convergent and $\lim_{k \rightarrow +\infty} A^k = \mathbb{0}_{n \times n}$

How to characterize semi-convergent and convergent matrices?

Characterization for symmetric A

Assume $A \in \mathbb{R}^{n \times n}$ $A = A^T$

Since symmetric, $\{\lambda_i\}_{i=1}^n \in \mathbb{R} \Rightarrow A$ can be diagonalized via orthogonal transf. $T \in \mathbb{R}^{n \times n}$ ($TT^T = T^T T = I$)

$$A = T \Lambda T^T$$

$$\text{Then } A^k = \underbrace{A \cdots A}_k = \underbrace{(T \Lambda T^T) (T \Lambda T^T) \cdots (T \Lambda T^T)}_k = T \Lambda^k T^T \Rightarrow$$

$$= T \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} T^T$$

$$\textcircled{1} A \text{ convergent} \Leftrightarrow |\lambda_i| < 1$$

$$\textcircled{2} A \text{ semi-conv} \Leftrightarrow |\lambda_i| \leq 1$$

• If $|\lambda_i| = 1 \Rightarrow \lambda_i = \pm 1$

Jordan normal form

Thm (Jordan normal form): For any matrix $A \in \mathbb{R}^{n \times n}$ there exists an invertible matrix $T \in \mathbb{R}^{n \times n}$ such that

$$J = T^{-1}AT = \begin{bmatrix} J_1 & 0 & \cdots & 0 \\ 0 & J_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & J_m \end{bmatrix} \in \mathbb{C}^{n \times n}, \text{ with } J_i = \begin{bmatrix} \lambda_i & 1 & \cdots & 0 \\ 0 & \lambda_i & \ddots & 0 \\ \vdots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & \lambda_i \end{bmatrix} \in \mathbb{C}^{j_i \times j_i}.$$

▷ $\{\lambda_i\}_{i=1}^m$ are the eigenvalues (not necessarily distinct) of A

▷ algebraic multiplicity (a.m.) of λ_i = sum of sizes of all Jordan blocks with eig. λ_i

▷ geometric multiplicity (g.m.) of λ_i = number of Jordan blocks with eig. λ_i

▷ λ_i is simple if a.m. = g.m. = 1, semisimple if a.m. = g.m. ↪ # of indep eigvect of λ_i

Example: $A \in \mathbb{R}^{9 \times 9}$

$$J = \begin{bmatrix} \boxed{\begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix}} & & & & \\ & \boxed{2} & & & \\ & & \boxed{\begin{matrix} 3 & 1 \\ 1 & 3 \end{matrix}} & & \\ & & & \boxed{7} & \\ & & & & \boxed{4} \\ & & & & & \boxed{4} \end{bmatrix}$$

$$\lambda_1 = 1 \quad \text{a.m.} = 4 \quad \text{g.m.} = 2$$

$$\lambda_2 = 3 \quad \text{a.m.} = 2 \quad \text{g.m.} = 1$$

$$\lambda_3 = 7 \quad \text{a.m.} = 1 = \text{g.m. (simple)}$$

$$\lambda_4 = 4 \quad \text{a.m.} = 2 = \text{g.m. (semi-simple)}$$

Characterization of convergent and semi-convergent matrices

For a matrix $A \in \mathbb{R}^{n \times n}$ with Jordan blocks $\{J_i\}_{i=1}^m$:

$$A^k = T J^k T^{-1} = T \begin{bmatrix} J_1^k & 0 & \dots & 0 \\ 0 & J_2^k & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & J_m^k \end{bmatrix} T^{-1}, \quad J_i^k = \begin{bmatrix} \lambda_i^k & \binom{k}{1} \lambda_i^{k-1} & \dots & \binom{k}{j_i-1} \lambda_i^{k-j_i+1} \\ 0 & \lambda_i^k & \ddots & \vdots \\ \vdots & \ddots & \ddots & \binom{k}{1} \lambda_i^{k-1} \\ 0 & \dots & 0 & \lambda_i^k \end{bmatrix}$$

A (semi-)convergent $\iff$ all J_i (semi-)convergent

Thm (Convergence and spectral radius): Consider a matrix $A \in \mathbb{R}^{n \times n}$ and let $\rho(A) := \max\{|\lambda| \mid \lambda \in \text{spec}(A)\}$ (spectral radius of A). Then:

- (i) A convergent $\iff \rho(A) < 1$;
- (ii) A semi-convergent and not convergent $\iff 1$ is a semisimple eigenvalue of A and all the other eigenvalues have modulus strictly less than 1.

Note: $\binom{k}{m} = \frac{k!}{m!(k-m)!} \leq \frac{k^m}{m!}$

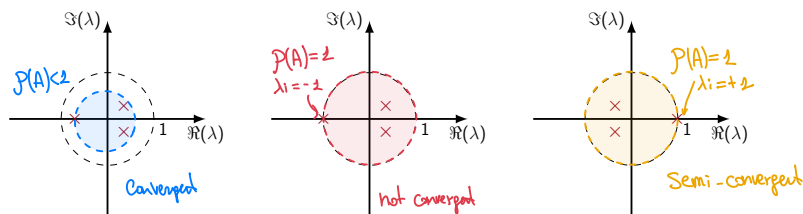
$\lim_{k \rightarrow \infty} \binom{k}{m} \lambda_i^{k-m}$ If $|\lambda_i| < 1$: $\left| \binom{k}{m} \lambda_i^{k-m} \right| = \binom{k}{m} |\lambda_i|^{k-m} \leq \frac{k^m}{| \lambda_i |^m m!} |\lambda_i|^k \rightarrow 0$

If $\lambda_i = 1, m=0$: $\binom{k}{0} 1^k = 1$

Otherwise $\lim_{k \rightarrow \infty} \binom{k}{m} \lambda_i^{k-m}$ doesn't exist finite

Characterization of convergent and semi-convergent matrices

Q: $A \in \mathbb{R}^{3 \times 3}$ with the following spectrum is (semi-)convergent?



Stochastic matrices → Describe averaging systems

Def: A matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ is

- (i) non-negative (positive) if $a_{ij} \geq 0$ ($a_{ij} > 0$, resp.) for all $i, j \in \{1, \dots, n\}$;
- (ii) row-stochastic if it is non-negative and $A\mathbb{1}_n = \mathbb{1}_n$;
- (iii) column-stochastic if it is non-negative and $A^\top \mathbb{1}_n = \mathbb{1}_n$;
- (iv) doubly-stochastic if it is row- and column-stochastic.

▷ If A is non-negative (positive) we will write $A \geq 0$ ($A > 0$, resp.)

Q: A is row- / column- / doubly-stochastic?

$$A = \begin{bmatrix} 1 & 0 & -0.5 \\ 0 & 0.5 & 0 \\ 0 & 0.5 & 1.5 \end{bmatrix}, \quad A = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.6 & 0.1 & 0.3 \\ 0.4 & 0.4 & 0.2 \end{bmatrix}, \quad A = \frac{1}{15} \begin{bmatrix} 4 & 9 & 2 \\ 3 & 5 & 7 \\ 8 & 1 & 6 \end{bmatrix}$$

↪ A not non-negative
↪ row-stoch
↪ stoch²

Properties of stochastic matrices

Lemma (Geršgorin disks): For any matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$,

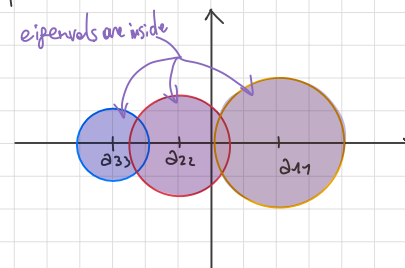
$$\text{spec}(A) \subseteq \bigcup_{i \in \{1, \dots, n\}} \left\{ z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{j=1, j \neq i}^n |a_{ij}| \right\}$$

Thm (Spectral properties of row-stochastic matrices): For a row-stochastic matrix $A \in \mathbb{R}^{n \times n}$

- (i) $1 \in \text{spec}(A)$;
- (ii) $\text{spec}(A)$ is a subset of the unit disk and $\rho(A) = 1$.

$$\{ z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{j=2, j \neq i}^n |a_{ij}| \} \equiv i\text{-th disk}$$

eigenvals are inside
circle with center a_{ii} and radius $\sum_{j \neq i} |a_{ij}|$

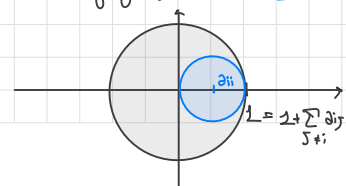


Proof: A row-stoch, then $A\mathbb{1}_n = \mathbb{1}_n$, hence $\mathbb{1}_n$ eig-vec of A with eigenval 1. This also implies $\rho(A) \geq 1$

To prove that $\rho(A) = 1$, we show that $\rho(A) \leq 1$ using Geršgorin disk th (Recall: i -th disk $D_i = \{ z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \}$)

Since A row-stoch: 1) $0 \leq a_{ii} \leq 1 \quad \forall i$

$$2) a_{ii} + \sum_{j \neq i} a_{ij} = 1 \quad \forall i$$



1) and 2) imply that the center of each D_i is $[0, 1]$ and the right-most point of D_i is 1. Then all D_i are subsets of unit disk and so it's their union. Thus by Geršgorin disk $\text{spec}(A)$ is a subset of unit disk and $\rho(A) \leq 1$

Other special classes of non-negative matrices

Def: A non-negative matrix $A \in \mathbb{R}^{n \times n}$, $n \geq 2$, is

- (i) irreducible if $\sum_{k=0}^{n-1} A^k$ is a positive matrix;
- (ii) primitive if there exists $k \in \mathbb{N}$ such that A^k is positive;
- (iii) reducible if it is not irreducible.

Q: A is irreducible / primitive?

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

↓
 $\tilde{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Note: 1) A is irreducible $\Leftrightarrow \forall i, j \exists k_{ij} \in \mathbb{N}$ st $[A^{k_{ij}}]_{ij} > 0$

2) If A is primitive, then the smallest K st $A^K > 0$ is called primitivity index of A . It can be proved that the primitivity index is always $\leq (n-1)^2 + 1$

3) If $\exists k \in \mathbb{N}$ st $A^k > 0 \Rightarrow A^{k+1} > 0, A^{k+2} > 0 \dots$

1) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ irr since $A^0 + A^1 + A^2 = \text{ones}(3,3) > 0$ (but not primitive)

2) $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $A^2 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $A^3 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ $A^4 = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ $A^5 = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} > 0$
primitive

$A^0 + A + A^2 > 0$ irr

3) $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{bmatrix}$ REDUCIBLE since $A^k = \begin{bmatrix} \bar{A}^k & 0 \\ * & 2 \end{bmatrix}$

Other special classes of non-negative matrices

Lemma: If a non-negative matrix is primitive, then it is also irreducible.

non-negative

irreducible

primitive

positive

Perron-Frobenius theory

Thm (Perron-Frobenius): Consider a matrix $A \in \mathbb{R}^{n \times n}$. If A is non-negative, then

- (i) there exists a real eigenvalue $\lambda \geq |\mu| \geq 0$ for all other eigenvalues μ ;
- (ii) the right and left eigenvectors v and w of λ can be selected non-negative.

If additionally A is irreducible, then

- (iii) the eigenvalue λ is strictly positive and simple; $\rightarrow \text{alg mult} = \text{geom mult} = 1$
- (iv) the right and left eigenvectors v and w of λ can be selected positive.

If additionally A is primitive, then

- (v) the eigenvalue λ satisfies $\lambda > |\mu|$ for all other eigenvalues μ .

▷ $\lambda = \rho(A)$ and λ is called dominant eigenvalue of A

▷ v and w are called dominant right and left eigenvectors of A

Perron-Frobenius theory

Q: connect each matrix to its spectrum

IRR + NOT PRIM

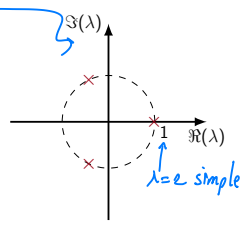
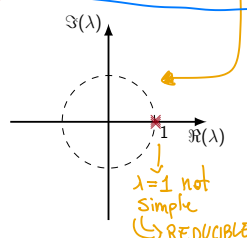
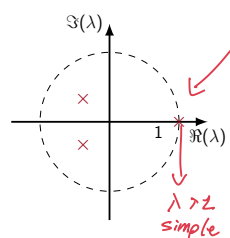
$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

PRIM

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

REDUCIBLE

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$



Note: There exist reducible matrices for which the dominant eigenvalue λ is strictly positive

and $\lambda > |\mu|$ for all other eigs μ

For example $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$ eig $\lambda_1=3$ $\lambda_2=0.5$ $\lambda_3=0$
and the matrix is reducible

Applications to DT averaging sytems

Consider the DT linear averaging system:

$$x(k+1) = Ax(k), \quad k \in \mathbb{Z}_{\geq 0}, \quad x(0) = x_0 \in \mathbb{R}^n, \quad A \text{ row-stochastic}$$

whose solution is

$$x(k) = A^k x_0, \quad k \in \mathbb{Z}_{\geq 0}.$$

If A is primitive then by PF theory, A is semi-convergent, so $\lim_{k \rightarrow \infty} x(k)$ exists finite.

Handwritten notes:
 $\rightarrow \text{row-stoch} \Rightarrow \rho(A) = 1$
 $\rightarrow \text{Prim} \Rightarrow \lambda = \rho(A)$ simple and all other eigvals μ are s.t. $|\mu| < 1$

To which value does $x(k)$ converge to?

Recall

Def: A matrix $A \in \mathbb{R}^{n \times n}$ is

- (i) semi-convergent if $\lim_{k \rightarrow +\infty} A^k$ exists (finite);
- (ii) convergent (or Schur stable) if it is semi-convergent and $\lim_{k \rightarrow +\infty} A^k = 0_{n \times n}$

$$\textcircled{1} A \text{ convergent} \Leftrightarrow |\lambda_i| < 1$$

$$\textcircled{2} A \text{ semi-conv} \Leftrightarrow |\lambda_i| \leq 1$$

$$\cdot \text{ If } |\lambda_i| = 1 \Rightarrow \lambda_i = +1$$

Thm (Convergence and spectral radius): Consider a matrix $A \in \mathbb{R}^{n \times n}$ and let $\rho(A) := \max\{|\lambda| \mid \lambda \in \text{spec}(A)\}$ (spectral radius of A). Then:

- (i) A convergent $\Leftrightarrow \rho(A) < 1$;
- (ii) A semi-convergent and not convergent $\Leftrightarrow 1$ is a semisimple eigenvalue of A and all the other eigenvalues have modulus strictly less than 1.

Applications to DT averaging sytems

Thm (Powers of row-stochastic matrices with a simple and strictly dominant eigenvalue): Consider a row-stochastic matrix $A \in \mathbb{R}^{n \times n}$ and assume that its eigenvalue 1 is simple and strictly larger in modulus than all other eigenvalues. Then A is semi-convergent and

$$\lim_{k \rightarrow +\infty} A^k = \mathbb{1}_n w^T$$

where $w \geq 0$ is the left dominant eigenvector of A normalized so that $w^T \mathbb{1}_n = 1$.

- ▷ If A is row-stochastic and primitive then the assumption is automatically satisfied!
- ▷ $\mathbb{1}_n w^T$ is a rank-1 projection matrix.

A is a projection matrix if $A^2 = A \longrightarrow \mathbb{1}_n (\overset{=1}{W^T} \mathbb{1}_n) W^T = \mathbb{1}_n W^T$

Proof: Since 1 is a simple eigval of A , $\exists T \in \mathbb{R}^{n \times n}$ inv. s.t. $A = T \begin{bmatrix} 1 & 0 \\ 0 & \bar{A} \end{bmatrix} T^{-1}$, $\bar{A} \in \mathbb{R}^{(n-2) \times (n-2)}$ *J. canonical form*

Since 1 is strictly dominant! (that is, all other eigenvalues have magnitude strictly smaller than 1),

then $\rho(\bar{A}) < 1$. So $\bar{A}$ is convergent ($\lim_{k \rightarrow \infty} \bar{A}^k = 0$)

$$\text{Then } \lim_{k \rightarrow \infty} A^k = \lim_{k \rightarrow \infty} T \begin{bmatrix} 1 & 0 \\ 0 & \bar{A}^k \end{bmatrix} T^{-1} = T \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} T^{-1} = v_2 w_2^T \text{ where } T = [v_2 \dots v_n], T^{-1} = \begin{bmatrix} w_2^T \\ \vdots \\ w_n^T \end{bmatrix}$$

Note that v_2 and w_2 are the right and left dominant eigenvectors of A , in fact $A = T J T^{-1} \Rightarrow A T = T J$

$$\text{Hence } A [v_2 \dots v_n] = [v_2 \dots v_n] \begin{bmatrix} 1 & 0 \\ 0 & \bar{A} \end{bmatrix} \xrightarrow{\text{(first col)}} A v_2$$

Applications to DT averaging systems

Thm (Powers of row-stochastic matrices with a simple and strictly dominant eigenvalue): Consider a row-stochastic matrix $A \in \mathbb{R}^{n \times n}$ and assume that its eigenvalue 1 is simple and strictly larger in modulus than all other eigenvalues. Then A is semi-convergent and

$$\lim_{k \rightarrow +\infty} A^k = \mathbb{1}_n w^\top$$

where $w \geq 0$ is the left dominant eigenvector of A normalized so that $w^\top \mathbb{1}_n = 1$.

- ▷ If A is row-stochastic and primitive then the assumption is automatically satisfied!
- ▷ $\mathbb{1}_n w^\top$ is a rank-1 projection matrix.

Similarly: $A = T J T^{-1} \Rightarrow T^{-1} A = J T^{-1} \Rightarrow \begin{bmatrix} w_2^\top \\ \vdots \\ w_n^\top \end{bmatrix} A = \begin{bmatrix} 1 & 0 \\ 0 & \bar{A} \end{bmatrix} \begin{bmatrix} w_2^\top \\ \vdots \\ w_n^\top \end{bmatrix}$

$\xRightarrow{\text{(first row)}} w_2^\top A = w_2^\top$

Moreover since A is row-stochastic, $v_2 = \alpha \mathbb{1}_n$, $\alpha \in \mathbb{R}$, $\alpha \neq 0$

We can choose $\alpha = 1$ and from $T^{-1} T = \mathbb{I}_n$

$$= \begin{bmatrix} w_2^\top \\ \vdots \\ w_n^\top \end{bmatrix} \begin{bmatrix} v_2 & \dots & v_n \end{bmatrix} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 2 \end{bmatrix} \Rightarrow w_2^\top v_2 = 1$$

$\mathbb{1}_n$

Applications to DT averaging sytems: example

▷ sensors $\{1, 2, \dots, n\}$

▷ $\theta \in \mathbb{R}$ unknown parameter

▷ local measurement: $y_i = \theta + v_i$,
 $v_i \sim \mathcal{N}(0, \sigma^2)$, $v_i \perp v_j$, $i \neq j$

▷ least-square estimate: $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n y_i$

Centralized
Computation
can be problematic
for n large



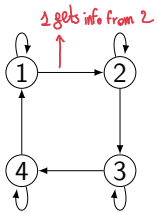
Distributed averaging algorithm

▷ Initialize $x_i(0) = y_i$ and then repeat for all $k \geq 0$

$$x_i(k+1) = \sum_{j \in \text{Neighbors}(i)} a_{ij} x_j(k), \quad a_{ij} \geq 0, \quad \sum_{j=1}^n a_{ij} = \sum_{i=1}^n a_{ij} = 1$$

By averaging with neighbors we reduce the computational burden

Applications to DT averaging sytems: example



$$A = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0 & 0.5 \end{bmatrix}, \quad x(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \end{bmatrix}, \quad x(0) = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

A doubly-stochastic + primitive $\implies w = \frac{1}{4} \mathbf{1}_4 \implies \lim_{k \rightarrow \infty} A^k = \frac{1}{4} \mathbf{1}_4 \mathbf{1}_4^\top$

Simple & strictly dominant eigval in 1

$$\lim_{k \rightarrow \infty} x(k) = \lim_{k \rightarrow \infty} A^k y = \frac{1}{4} \mathbf{1}_4 \mathbf{1}_4^\top y = \left(\frac{1}{4} \sum_{i=1}^4 y_i \right) \mathbf{1}_4$$

→ from prev TH. I know that $A^k \rightarrow \mathbf{1} \mathbf{w}^\top$ (Recall: $\mathbf{w}^\top \mathbf{1} = 1$)

All the nodes will converge to such quantity
 i.e. local computations